VEHICLE HEALTH ANALYSIS AND SPARE PARTS E-COMMERCE

A PROJECT REPORT

Submitted by

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ABSTRACT

Vehicles need constant servicing and repairs after a certain interval of time and if not repaired at fixed intervals it can lead to accidents. The objective of the project is to give an idea to the owner about the estimated insurance claim which he/she can get in case of certain faults and issues whether the insurance claim exists for the vehicle or not. There are many automobile health monitoring systems which are used to detect and diagnose faults and issues like On-Board Diagnostic (OBD) and Engine Control Units (ECU) which tells the owner where the fault has occurred. In this project, we have used Support Vector Regression, TFIDF, LSTM and a dataset containing the car prices to train a machine learning algorithm that can predict the estimated insurance claim based on the faults which the owner has entered.

The main purpose of this system is to reduce the amount of scams that takes place when the owner has no idea about the actual cost. This project aims to mitigate it by helping the owner to know about their vehicle's full insurance which they can rightfully claim and the amount they are required to pay.

Keywords — On-Board Diagnostic, Engine Control Units, Support Vector Regression, TFIDF, LSTM

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Own Work Declaration Form*

Degree/ Course **: Btech/CSE-Big Data Analytics**

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Yuvraj Singh Chauhan

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CHAPTER 1

Introduction

Many of today's vehicles include a wide range of systems and components that perform various operations while the vehicle is in use. Over time, repeated use of the vehicle may cause failure of individual systems or components. The failure of one component or system affects the overall performance of a vehicle. For continuous better performance of a vehicle, the best way is to carry out prescribed periodical inspection and repair in a prescribed way. It requires years of experience to inspect manually, thus the electronic vehicle health monitoring system is used. In order to get the maximum insurance sum of the vehicle that falls under the company’s insurance claim policy, this is where vehicle insurance application comes which helps the owner to get the maximum possible claim.

Insurance systems are widespread in today's world. They are being extensively deployed to check and verify the insurance sum of a particular entity. To name a few, insurance systems are used in healthcare, business, home, life and travel. Insurance systems use the information of the various cost prices of the particular entities which they use along with the current issue in order to estimate the insurance claim. Vehicle Insurance Application uses machine learning algorithms like Supportvector regression, TFIDF and LSTM to predict an estimated price

of the insurance sum based on the faults given by the owner. Thus, it helps in reducing scams like not paying the money, giving way less money than the actual amount, etc., and helps the owners to get their maximum possible claim.

A contract between an insurer and an insured under which the insurer agrees to make up any losses resulting from the risk insured against is known as an insurance policy. In exchange, the insured agrees to pay an upfront amount known as the premium. The Policy is the name of the legal document comprising the insurance contract. The objects, such as property or lives, that serve as the foundation for insurance are referred to as the subject matter of insurance. The Legislature enacted the Insurance Act of 1938 on February 26, 1968, and it went into effect on July 1, 1939. It has undergone several revisions. It encompasses all of India. It is an act to alter and combine the laws governing the insurance industry.

In practice, insurance entails everyone in the pool agreeing to shoulder a percentage of the loss. Because it is unclear who will lose money in such a deal, everyone has found it to be favorable. Thus, throughout time, the notion that such a shared resource pool should be managed by professionals who would determine the amount of the contribution to be imposed on each individual emerged. The concept of insurance emerged in this manner. Although insurance has evolved into a highly commercial

endeavor in contemporary times, the basic idea has remained the same: the insurer collects premiums from a large number of people and insures them against a wide range of hazards.

Hurdles

• Dataset was a major issue for this research since many locations did not have adequately organised datasets for analysis. Some were marked with a name and monetary worth, while others showed only their total distance travelled. At some stage in this procedure, there was not enough information for the machine to function, and predictions were off.

• After much effort, we were able to locate an appropriate dataset for the study, one that included relevant information on car engines such as horsepower, mileage, cylinders, and strokes.

• It was feasible for linear regression to have negative findings, indicating that the vehicle was either not worth insuring or had poor health.

• Even though the value is predicted, more information on vehicle and previous insurance issued for those along with additional attributes can really help the machine provide more accurate value. • These cases occurred frequently during the data analysis and prediction time, which took a lot of time.

Additionally, additional information on this value prediction may be obtained by the use of different algorithms such as SVM, logistic regression, Decision tree, KNN, and K-mean.

CHAPTER 2

Literature Survey

I. GAFFAR GULAB MOMIN

"A REVIEW ON VEHICLE HEALTH ANALYSIS MONITORING SYSTEM" [2022]

After a given amount of time, vehicles require repair, and if such repairs are not made at regular intervals, accidents may occur. The project's goal is to continually alert the owner to issues such as tyre air pressure, exhaust gas quality, brake wear, oil leaks, excessive engine heating, fuel pipe blockages, etc. Tire pressure monitoring sensors are used to obtain real-time tyre pressure readings. Gas sensors provide readings for carbon dioxide (CO2), nitrogen oxides (NOx), and other gases, and the quantity and quality of exhaust determine the engine's proper operating condition. Flow sensors are used to detect oil overflows and leaks. Temperature sensors are used in engines to alert the owner when the engine is overheating.

II. E. CHANDRASEKHAR

"VEHICLE HEALTH MONITORING SYSTEM" [2021]

The phrase "on-board diagnostics'' in the automobile industry refers to a vehicle's capacity for self-diagnosis and reporting. The state of the different car subsystems may be accessed by the owner or a repair professional using OBD systems. Since its inception in the first iterations of on-board car computers in the early 1980s, the quantity of diagnostic

data accessible via OBD has changed greatly. When a fault was discovered, older versions of OBD would only turn on a "idiot light" or malfunction indicator light without giving any other details. Modern OBD implementations give real-time data via a standardized digital communications interface in addition to a defined set of diagnostic problem codes that enable quick identification and correction of vehicle issues. OBD-I or OBD-II buses are often included in cars to specify reports or diagnostics of the vehicle.

III. PARMESH K, RASHMI PRAFULLAKUMAR NERIYA, GANESAN K

"VEHICLE HEALTH MONITORING AND ANALYSIS" [2015]

Modern automobiles are more intelligent in terms of engine choices, performance, fuel efficiency, security, and stability thanks to the integration of a higher number of ECUs and sensors. Through the OBDII (on-board diagnostics) protocol, ECUs also offer helpful information in the form of diagnostic codes. These codes identify a particular problem with the vehicle's system and are used by service personnel to fix the problem. In order to monitor the vehicle's health in real-time, this article investigates the possibilities of OBDII codes from the engine ECU and computes the data on an Android tablet. The goal is to offer portable diagnosis solutions and incorporate such systems into in-car entertainment systems like Apple CarPlay and Google Android Auto, which can be essential elements for V2X and IOVs (Internet of vehicles).

IV. MING-SONG LIN, CUN-XIANG LIU

"RESEARCH OF THE SYSTEM OF REMOTING VEHICLE INSURANCE ESTIMATION" [2016]

The present approach for estimating auto insurance has a lot of issues and takes too long. With the help of the remote vehicle insurance estimation system, the insurance provider will receive a picture of the vehicle loss situation in order to assess the damage promptly, make a quick decision, expedite processing of claims, reduce operating costs, and enhance the effectiveness of managing compensation. In contrast, it is clear that the remote vehicle insurance estimation system, which is a long-range system as opposed to a vehicle insurance estimate model, is more suited to meet the demands of the market today. A significant promotion is required in the Chinese insurance industry.

V. UMA SHANKAR YADAV, RAVINDRA TRIPATHI, MANO ASHISH TRIPATHI

"REVIEW OF THE MOTOR VEHICLE INSURANCE INDUSTRY IN INDIA" [2022]

The most popular and efficient way to safeguard personal property is through motor vehicle insurance, which has garnered a lot of interest. Understanding the present state of the Malaysian auto insurance market is essential. This setting serves as the foundation for this article's analysis of the Indian motor insurance sector's growth model and its connection to the country's economic growth. This article examines methods for achieving targeted product advertising and regionally-specific motor vehicle insurance (VMI). The goal of this paper is to build a solid

foundation for fostering the growth of the Indian car insurance market and its connection to the country's vehicle manufacturing. The article examines the impact of Malaysia's VMI policy on the nation's economy and motor vehicle manufacturing in comparison to other developed nations. The article also emphasizes the impact of COVID-19 on the VMI sector.

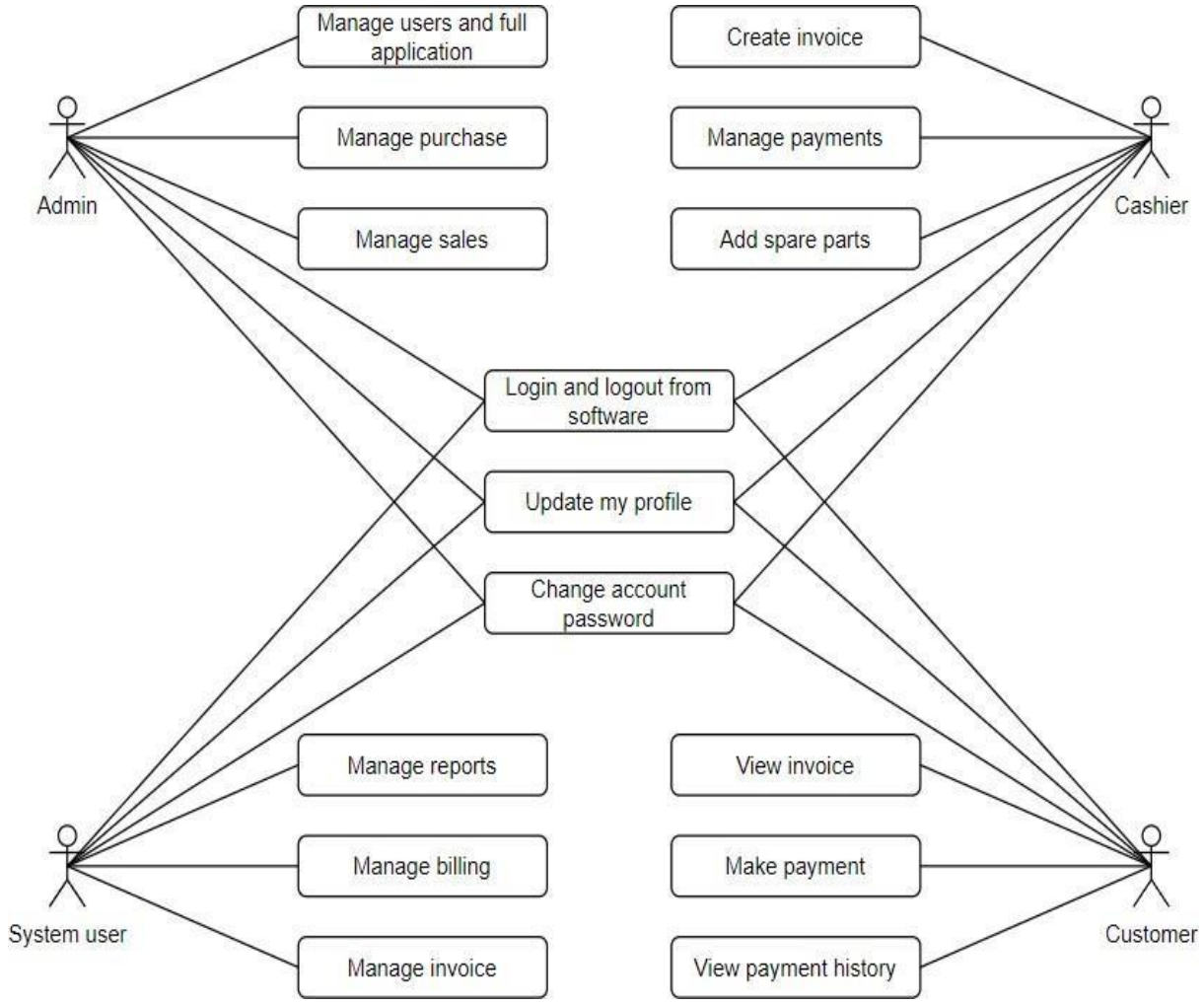
VI. JACOB MEHLMAN, JIAHUI WU, HANSHEN JING JINGYI QI

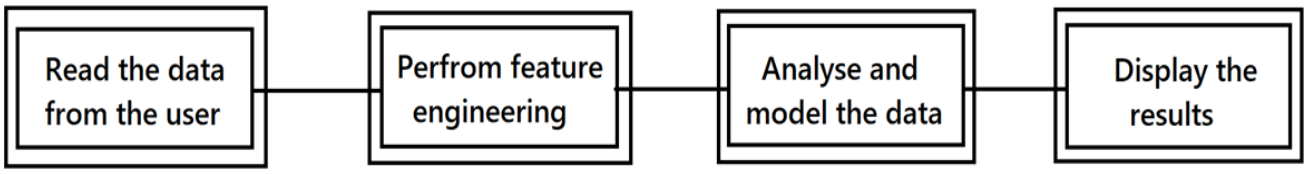
"PREDICTION OF CUSTOMER'S INTERESTS IN VEHICLE INSURANCE" [2022]

The insurance company that has provided health insurance to its customers now needs to find whether policyholders(customers) from the past year will also be interested in vehicle insurance provided by the company. In this study, we build statistical models using bagging, AdaBoost, random forest, XGBoost, and Convolutional Neural Network based on demographic information, vehicle situation, and insurance policy information to predict customers' willingness to buy vehicle insurance. We generate ROC curves, confusion matrices, and accuracy scores to compare the models and finally choose the CNN model and make recommendations to the insurance company.

CHAPTER 3

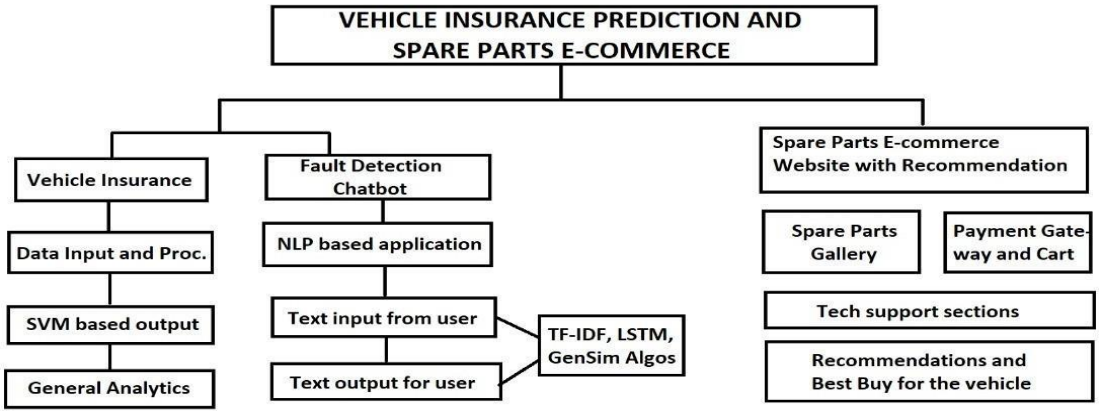
Block Diagram

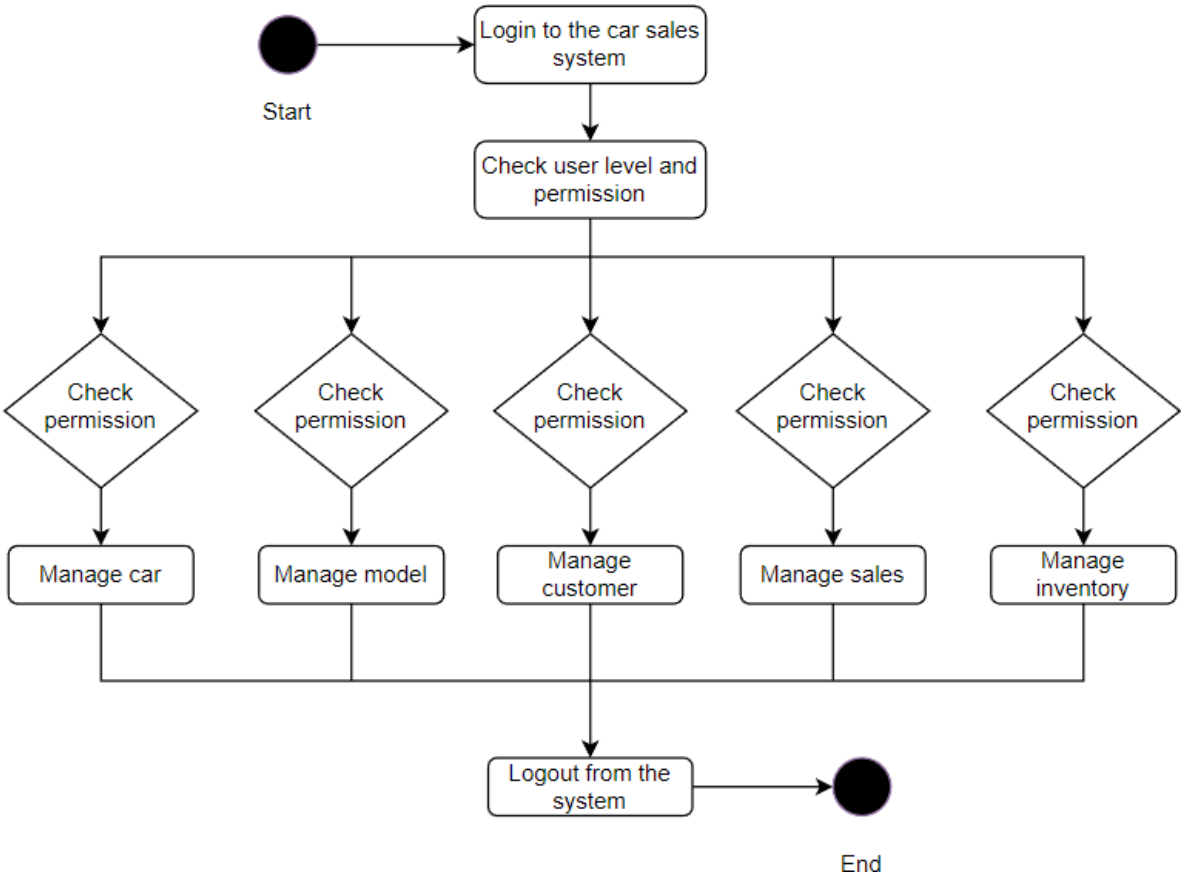


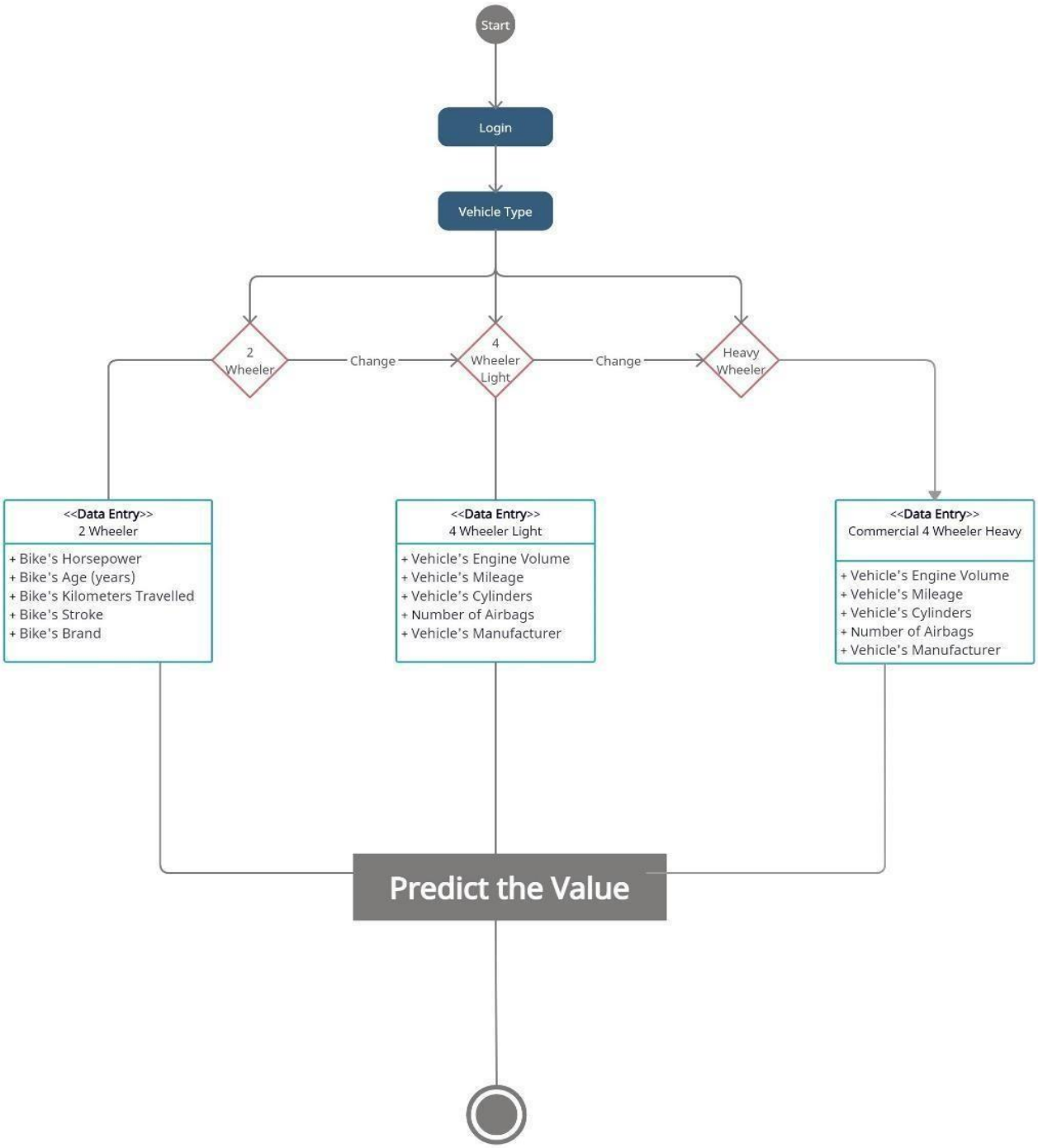


CHAPTER 4

UML Diagram






Start
Login
Vehicle Type
2 Wheeler
Change
4 Wheeler Light
Change
Heavy Wheeler
<<Data Entry>>
2 Wheeler
+ Bike's Horsepower
+ Bike's Age (years)
+ Bike's Kilometers Travelled
+ Bike's Stroke
+ Bike's Brand
<<Data Entry>>
4 Wheeler Light
+ Vehicle's Engine Volume
+ Vehicle's Mileage
+ Vehicle's Cylinders
+ Number of Airbags
+ Vehicle's Manufacturer
<<Data Entry>>
Commercial 4 Wheeler Heavy
+ Vehicle's Engine Volume
+ Vehicle's Mileage
+ Vehicle's Cylinders
+ Number of Airbags
+ Vehicle's Manufacturer
Predict the Value

CHAPTER 5

Algorithm

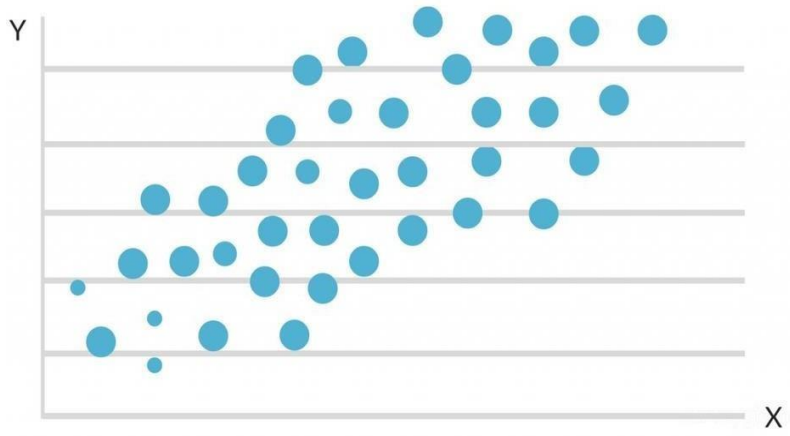
Finding the factors that have an effect on an interest issue may be done with accuracy using regression analysis. You can confidently establish which elements are most important, which ones can be ignored, and how these factors interact when you do a regression.

The following concepts must be understood in order to completely appreciate regression analysis:

Dependent Variable: This is the primary element you're attempting to comprehend or anticipate.

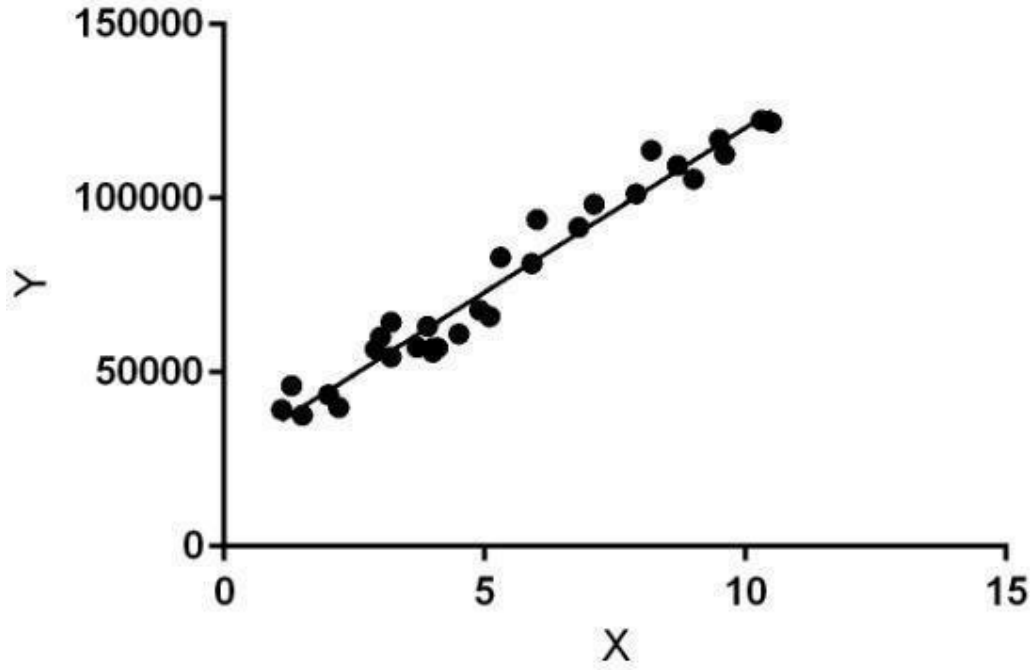
The variables that you assume have an effect on your dependent variable are known as independent variables.

The guests' level of pleasure with the event is our dependent variable in the aforementioned application training scenario. Our independent variables include the subjects addressed, session lengths, the amount of food offered, and the price of a ticket.



(1) Linear Regression

Linear regression is a supervised learning-based machine learning technique. A regression procedure is carried out. Regression models a desired prediction value using independent variables. It is mostly used to establish the relationship between variables and forecasts. Different regression models differ in the sort of relationship they consider between dependent and independent variables as well as the number of independent variables they use.



Using linear regression, one may forecast the value of a dependent variable (y) based on an independent variable (x). Therefore, using this regression technique, it is discovered that x (the input) and y (the result) are linearly connected (output). The phrase "linear regression" was therefore created.

In the illustration above, X stands for a person's employment history, and Y stands for their salary. The best fit for our model is the regression line.

Linear Regression Hypothesis Function:

$$y = \theta_1 + \theta_2.x$$

Input training data (one input variable, or parameter), x Labels to data, y As we train the given model (supervised learning)

During training, the model fits the line that predicts the value of y for a given value of x the best. The model generates the best regression fit line by identifying the best 1 and 2 values.

Interception is one.

x's coefficient is 2

The best 1 and 2 values are chosen, and then the best fit line is found. As a consequence, our model will predict the value of y given the value of x as an input whenever we use it to make predictions.

Cost Function (J): The model aims to anticipate y values with the minimal feasible error between the forecasts and the actual values by producing the best-fit regression line. It is essential to alter the values of 1 and 2 in

order to determine the best value that minimises the error between the predicted y value (pred) and real y value (y)

$$minimize \frac{1}{n} \sum_{i=1}^{n} (pred_i - y_i)^2$$

$$J = \frac{1}{n} \sum_{i=1}^{n} (pred_i - y_i)^2$$

The Root Mean Squared Error (RMSE) between the predicted y value (pred) and the real y value is the cost function(J) of linear regression (y). In the diagram above, X represents a person's job history and Y represents their wage. The regression line fits our model the best

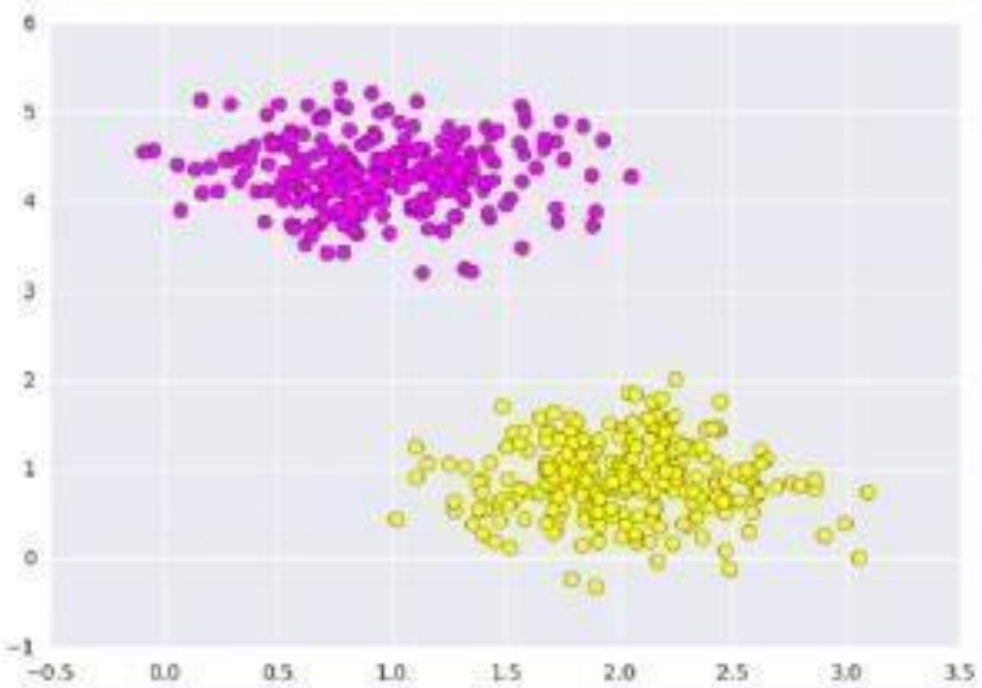
Cost function(J) of Linear Regression is the Root Mean Squared Error (RMSE) between predicted y value (pred) and true y value (y).

i. Support Vector Machine

Support vector machines (SVMs, or support vector networks) are supervised learning models in machine learning that employ associated learning methods to examine data for classification and

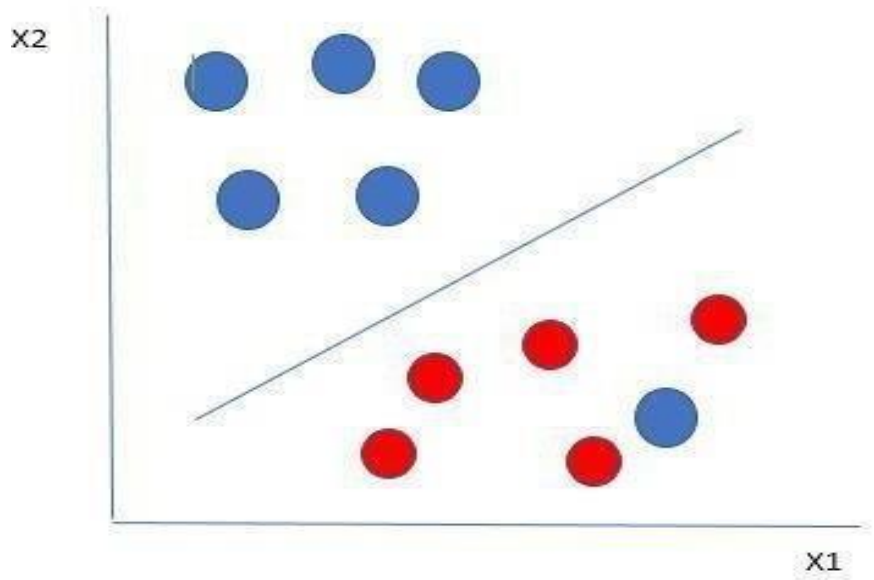
regression. Separating hyperplanes provide the formal definition for a Support Vector Machine (SVM), a discriminative classifier. The technique generates a hyperplane that best classifies fresh samples based on labelled training data (supervised learning).

In a support vector machine (SVM) model, the instances are represented as points in space, and the categories are mapped onto those points to create as large a gap as feasible between them. While SVMs were originally developed for linear classification, they have now been shown to be effective in non-linear classification as well, implicitly mapping their inputs into high-dimensional feature spaces.



Both classification and regression are accomplished with the help of Support Vector Machine (SVM), a supervised machine learning technique.

Although we commonly refer to regression problems, classification is the most suitable phrase. The SVM approach seeks to find a hyperplane in an N-dimensional space that cleanly categorises the data points. A hyperplane's dimension is proportional to the number of features. If only two variables are being used, the hyperplane will be very close to a straight line. If there are three input parameters, the hyperplane flattens down into two dimensions. Trying to picture something with more than three features is difficult.



Modules Used:

Module 1: Data Integration

Data from excel is integrated into the machine for data analysis. This data is then pre-processed using LabelEncoder(). Data of vehicles with attributes like engine volume, Mileage, Distance traveled, Horse Power,

Stokes, Airbags, Bike company are integrated into representing as a graph.

Module 2: **ML Algorithm**

Using python libraries, we integrate ML algorithms like SVM, Linear regression and many others to analyze the data. In this project we used only linear but other algorithms are also possible in which we can predict the values but making a slight change in the source code.

Module 3: **Value prediction and visualization**

In this module we have data visualization when graphs are being introduced to show relation between the attributes but plotting the graphs using matplotlib library. And using ML we predict the value to be paid as well. This prediction is selected by analysis of the data in the CSV files we integrated.

Among the Python libraries utilised is Streamlit, a free and open-source app framework. It speeds up the process of developing web applications for data science and machine learning.

Data science, data analysis, and machine learning are common applications of Pandas, a free and open-source Python tool.

If you're working with arrays in Python, you should check out the NumPy package. You may use its linear algebra, Fourier transform, and matrix operations tools.

• Matplotlib is a Python package for making all kinds of visualisations, from the static to the animated to the interactive.

Python's matplotlib is a feasible open-source alternative to MATLAB thanks to Pyplot, an API (Application Programming Interface).

• Sklearn contains numerous powerful methods for statistical modelling and machine learning, including dimensionality reduction, classification, regression, and clustering.

CHAPTER 6

Planning & Analysis

One of the most common types of non-life insurance is automobile insurance. This is due to the fact that it is legally required throughout the majority of the world. Every motor vehicle must be registered with the road transportation authority and have third-party liability insurance. The fundamental tenet is that a motor vehicle could result in property damage or bodily harm, necessitating insurance.

A legislative requirement under the Motor Vehicles Act is third-party liability coverage, generally known as motor third-party insurance or "act only" coverage. Since someone other than the two parties to the contract is the policy's beneficiary, it is known as a "third-party" cover (the car owner and the insurance company). The insured receives no benefits from the policy. It does, however, provide coverage for the insured's legal responsibility for loss or damage to third party property or death or disabling of a third party.

CHAPTER 7

Designing and Product Architecture

Analyze the information such as distance traveled since the time it is active, years used, model of car, and other factors to estimate the insurance amount to be paid and give maintenance alerts if required.

Give an option to the customer whether they like to order a technician to do the work or the customer themselves do it if they have the knowledge, however choosing the latter will not give any warranty or free replacement if anything goes wrong.

Estimate the insurance claim that would be received by the customer due to faults caused by some unforeseen circumstances.

The system will also provide e-commerce support where, if the issue is with a particular part, then that particular part will be searched online and displayed to the customer by using the NLP model.

CHAPTER 8

Implementation

Historically, technicians were called in order to assess the problems which the car has and they would suggest the estimated cost for fixing the issue as well.

In that system the chances of scamming the customers, not properly fixing the fault, not paying the insurance money to the customer due to lack of evidence and asking money way too much than the actual amount was common.

E-vehicle health monitoring and commerce systems were thus introduced to combat this problem from the market and provide a fair assessment for the problem at the best cost.

The current vehicle health monitoring systems uses ML algorithms to determine the condition of the vehicle and provide diagnosis for the same.

The replacement for the faulty parts is provided by the system at a minimum cost which is then installed by the technician.

If however the fault is there due to some unforeseen circumstances, diagnosis is done and the insurance money is paid to the customer.

CHAPTER 9

Developing

Following along with the developmental model fitting for this project, webpage representation was deemed a good fit for prototyping and representing the key functionalities of the product and its corresponding modules. HTML, CSS, Python and SQL languages have been used for making the frontend and the backend.

In addition to this, machine learning algorithms like Support Vector Machines and Linear Regression have been used in order to perform insurance prediction for the vehicle. In order to make the web application stream it was used.

With the use of module and a third party deployer, it was made possible for us to deploy a basic functioning website to analysis inputs and provide output for given data

CHAPTER 10

Testing

Locating errors is the main goal of testing. In order to ensure that the code works as expected, testers run it in different environments and analyse it in detail. Verification and validation are at the heart of the testing process, with the ultimate goal of identifying issues and figuring out how to repair them. It performs checks and tests on objects, such as software, performance, and consistency, and compares the outcomes to set criteria. Validation is done to ensure that all of the systems are functioning properly, while verification checks to determine if the software is performing as intended. When doing tests, it is useful to revisit the specifications for crucial quality features. All sorts of things are done during the testing process.

These include: 32 Analysis of Tests, Design of Tests, and Running of Tests.

It is essential to carry out these steps in order to cut down on costly and time-consuming rework. It is not possible to pause the software testing procedure in the middle of its progress.

For the following reasons, testing is an integral part of the software development life cycle (SDLC): • to find bugs • to clear up confusion • to boost brand image • to enhance product quality • to eliminate potential safety risks • to verify and validate design decisions • to reduce costs without sacrificing quality; • to increase usability; and • to broaden accessibility.

CHAPTER 11

Maintenance

It entails setting up the website on the computer system and starting the main operations. In this stage, users are introduced to the website for the first time and given the opportunity to interact with it. It entails keeping the website constantly updated to make sure it complies with the most recent updates to the same and current information needs. The developer evaluates the modifications and incorporates them as well. The following tasks are included in the implementation phase. the creation of websites, limiting which people may make changes to the website, and tracking logs to maintain tabs on data and website activity.

The WEB DEVELOPMENT LIFE CYCLE includes these phases, which take into account the following 10 guidelines:

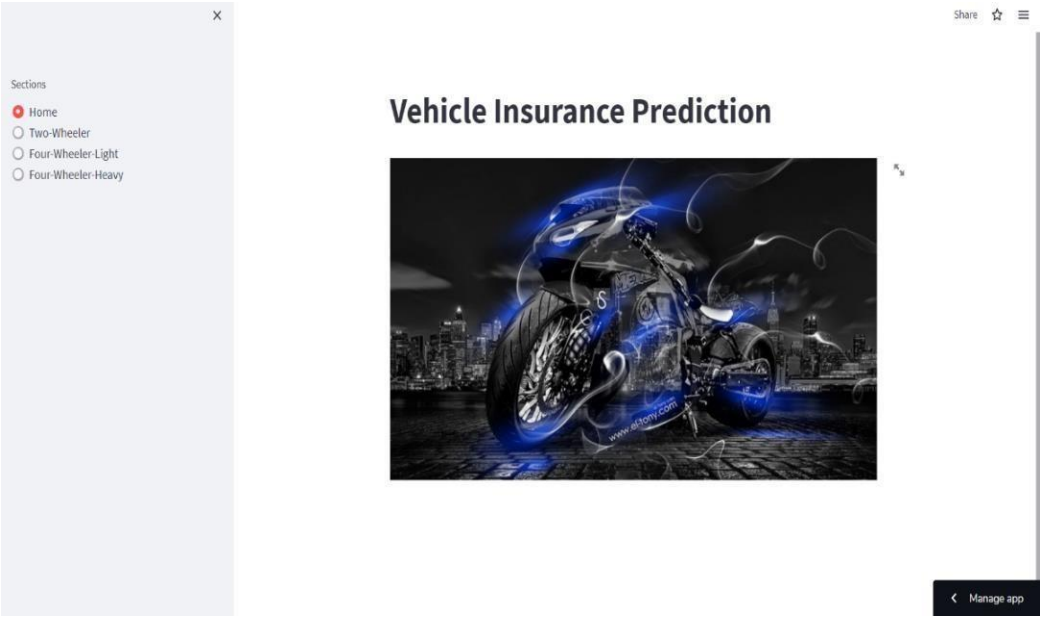
- Rule No. 1: Website Goals To get the intended results, the primary goal for the website's construction must be determined and evaluated.
- Rule 2: Website Map Prepare the overall website's physical look as well as its modularized structure. This might serve as a blueprint for future work.
- Rule 3: Gathering Requirements Include all client requirements and requirements criteria. The development process would be more succinct the more explicit the requirements are.
- Rule 4: Verification Before launching the website, the needs that

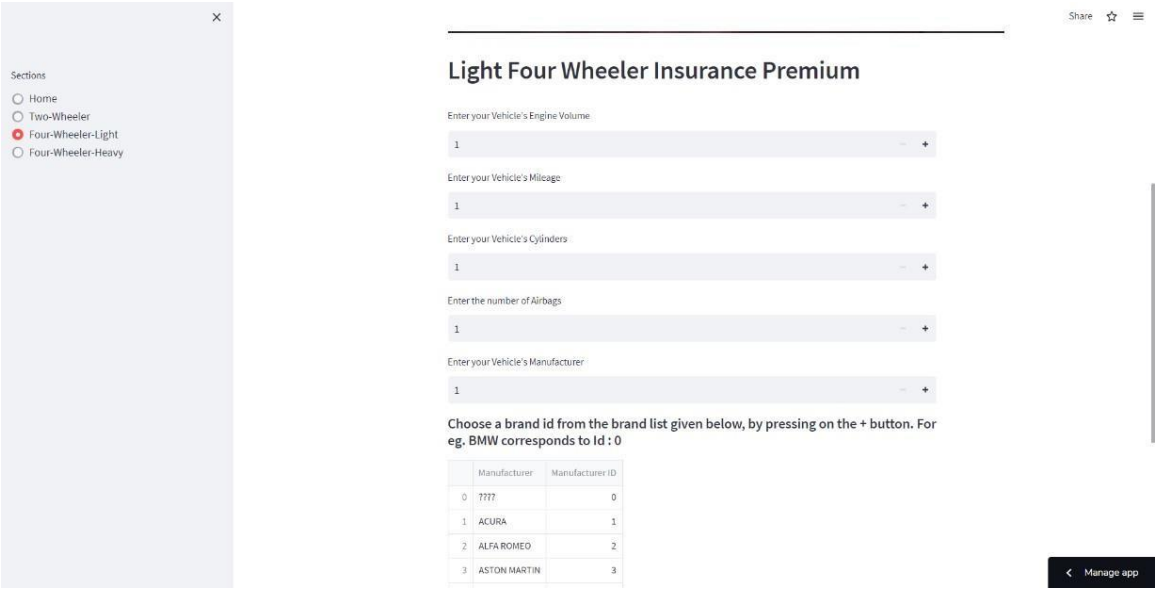
were gathered and the design criteria needed to be checked against intellectual property laws to prevent privacy concerns.

- Rule 5: Website designs should be straightforward, clear of ambiguity, and uncomplicated so that the general public may readily understand the material.
- Rule 6: Avoid shortcuts Offer readers navigational tools to make it easier for them to browse the website quickly and get the information they need.
- Rule 7: Unification After the structure is created, the website may be evaluated against other similarly functional websites to provide the user a special application experience.
- Rule 8: Multilingualism The extra features can be added to give multilingual, handicapped consumers with technical help.
- Rule 9: Testing After doing multiple rounds of testing, the developed web application should be published. The testing verifies that the completed system meets the initial objectives set out.
- Rule 10: Upkeep The website must be updated often to keep it accurate and of high caliber. Additionally, the consumer may receive current information from this.

CHAPTER 12

Screenshots





Two Wheeler Insurance Premium

Enter your Bike's Horsepower

200

Enter your Bike's Age (years)

2

Enter your Bike's Kilometers Travelled (in 1000 km)

2

Enter your Bike's Stroke

2

Enter your Bike's Brand

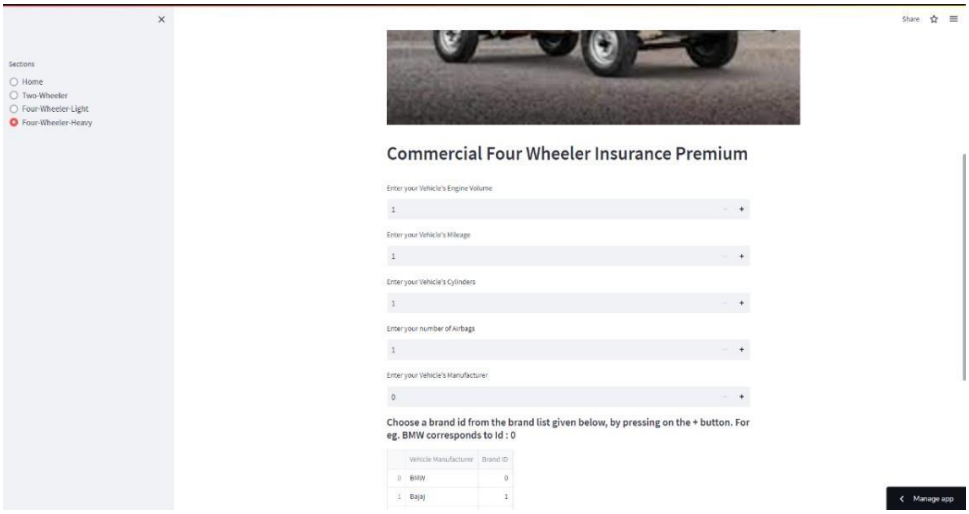
12

Choose a brand id from the brand list given below, by pressing on the + button. For eg. BMW corresponds to Id : 0

	Bike Brand	Brand ID
0	BMW	0
1	Bajaj	1
2	Benelli	2
3	Ducati	3
4	Harley-Davidson	4
5	Hero	5
6	Honda	6
7	Hyosung	7
8	Ideal	8
9	Indian	9

Predict

Your predicted premium price is Rs. 7833.1



130 lines (101 sloc) | 5.08 KB

```python
from numpy.core.fromnumeric import size
import streamlit as st
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn import preprocessing

st.title("Vehicle Insurance Prediction")



nav = st.sidebar.radio("Sections",["Home","Two-Wheeler","Four-Wheeler-Light","Four-Wheeler-Heavy"])
if nav == "Home":
    st.image("bike.jpg",width= 700)




elif nav == "Two-Wheeler":

    data = pd.read_csv("bikes.csv")
    data1 = data.copy()
    label_encoder = preprocessing.LabelEncoder()
    data['brand']= label_encoder.fit_transform(data['brand'])
    data['brand'].unique()
    lr = LinearRegression()
    Z = data[['power', 'age', 'kms_driven','stroke','brand']]
    lr.fit(Z, data['price'])
    st.image("bike.jpg",width= 800)
    st.header("Two Wheeler Insurance Premium")
    val1 = st.number_input("Enter your Bike's Horsepower",100,2000,step = 100)
    val2 = st.number_input("Enter your Bike's Age (years)",1,40,step = 1)
    val3 = st.number_input("Enter your Bike's Kilometers Travelled (in 1000 km)",1,50,step = 1)
    val4 = st.number_input("Enter your Bike's Stroke",1,5,step = 1)
    val5 = st.number_input("Enter your Bike's Brand",0,22,step = 1)

    val = np.array([val1,val2,val3,val4,val5]).reshape(1,-1)
```

```python
    pred = lr.predict(val)[0]
    pred = round(pred)/(10)
    if st.button("Predict"):
        st.success(f"Your predicted premium price is Rs. {pred}")


elif nav == "Four-Wheeler-Light":

    data = pd.read_csv("car_price_prediction.csv")
    data1 = data.copy()
    label_encoder = preprocessing.LabelEncoder()
    data['Manufacturer']= label_encoder.fit_transform(data['Manufacturer'])
    data['Manufacturer'].unique()
    lr = LinearRegression()
    Z = data[['Engine volume','Milage','Cylinders','Airbags','Manufacturer']]
    lr.fit(Z, data['Price'])
    st.image("car.jpeg",width= 800)
    st.header("Light Four Wheeler Insurance Premium")
    val1 = st.number_input("Enter your Vehicle's Engine Volume",1,20,step = 1)
    val2 = st.number_input("Enter your Vehicle's Mileage",1,100,step = 1)
    val3 = st.number_input("Enter your Vehicle's Cylinders",1,8,step = 1)
    val4 = st.number_input("Enter the number of Airbags",1,20,step=1)
    val5 = st.number_input("Enter your Vehicle's Manufacturer",1,63,step = 1)

    val = np.array([val1,val2,val3,val4,val5]).reshape(1,-1)

    datax = [data1["Manufacturer"], data["Manufacturer"]]
    headers = ["data1", "data"]
    df3 = pd.concat(datax, axis=1, keys=headers)
    dfx = df3.sort_values('data')
    dfx = dfx.drop_duplicates()
    dfx.reset_index(drop=True, inplace=True)
    st.markdown("##### Choose a brand id from the brand list given below, by pressing on the + button. For eg. BMW corresponds to Id : 0")
    dfx.rename(columns={'data1': 'Manufacturer','data':'Manufacturer ID'}, inplace=True)
    st.dataframe(dfx)


    pred = lr.predict(val)[0]
    pred = round(pred)
    if st.button("Predict"):
        st.success(f"Your predicted premium price is Rs. {pred}")
```

Light Four Wheeler Insurance Premium

Enter your Vehicle's Engine Volume

3

Enter your Vehicle's Mileage

25

Enter your Vehicle's Cylinders

4

Enter the number of Airbags

6

Enter your Vehicle's Manufacturer

7

Choose a brand id from the brand list given below, by pressing on the + button. For eg. BMW corresponds to Id : 0

	Manufacturer	Manufacturer ID
0	????	0
1	ACURA	1
2	ALFA ROMEO	2
3	ASTON MARTIN	3
4	AUDI	4
5	BENTLEY	5
6	BMW	6
7	BUICK	7
8	CADILLAC	8
9	CHEVROLET	9

Predict

Your predicted premium price is Rs. 34832

Links-

- https://vehicleinsurancepredictor18vk.streamlit.app
- https://bikevaluepredictorvk18.streamlit.app

CHAPTER 13

Result & Discussion

Dataset for this project was a big problem as many sites did not have properly arranged datasets for analysis. Some had name and value while some had distance traveled but no other attribute. In this process there was a point where insufficient data was available for the machine to work, wherein prediction was not as expected.

After a thorough search and time, we could find a dataset that could work with the analysis where it had horsepower, milage, stokes and other import elements for a vehicle's engine

When it comes to results, with the use of linear regression it was always possible for negative values which implies they are not worth insurance or the vehicle health was bad.

These cases occurred frequently during the data analysis and prediction time which took a lot of time

Even though the value is predicted, with more data on vehicle and previous insurance issued for those along with extra attributes can really help the machine provide more accurate value

Also, use of other algorithms like SVM, logistic regression, Decision tree, KNN, K-mean

can provide more insights on this prediction of value.

Conclusion

The vehicle monitoring system offers accurate information on all fundamental factors influencing a vehicle's performance. Due to the early detection of damage made possible by this information, the performance of the vehicle will improve. As a result, it gives the owner accurate information so that during car servicing, the service centers cannot present incorrect information and demand additional payment. The vehicle insurance app aids the owner in obtaining the most insurance payout possible in the event that the vehicle develops flaws or malfunctions as a result of normal wear and tear. According to the company's insurance policy, the owner can obtain the greatest claim possible, and that too without deceit or fraud. This method of auto insurance aids in eradicating such offensive behaviors.

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  Conference Publication
  Publication process not yet started

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7	Title of the Synopsis/ Thesis/ Dissertation/Project	Vehicle Health Analysis and Spare Sparts E-Commerce
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HH

PAGE 1

Prep. You may be using the wrong preposition.

PAGE 2

Article Error You may need to use an article before this word.

Article Error You may need to use an article before this word.

PAGE 3

Article Error You may need to remove this article.

PAGE 4

Sp. This word is misspelled. Use a dictionary or spellchecker when you proofread your work.

Sp. This word is misspelled. Use a dictionary or spellchecker when you proofread your work.

Article Error You may need to use an article before this word. Consider using the article **the**.

Run-on This sentence may be a run-on sentence.

PAGE 5

Confused You have used either an imprecise word or an incorrect word.

PAGE 6

Article Error You may need to remove this article.

P/V You have used the passive voice in this sentence. You may want to revise it using the active voice.

Article Error You may need to use an article before this word. Consider using the article **the**.

PAGE 7

Article Error You may need to use an article before this word.

PAGE 8

Article Error You may need to use an article before this word.

PAGE 9

Verb This verb may be incorrect. Proofread the sentence to make sure you have used the correct form of the verb.

PAGE 10

PAGE 11

Missing "," Review the rules for using punctuation marks.

Article Error You may need to remove this article.

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Compound These two words should be a compound word.

Run-on This sentence may be a run-on sentence.

PAGE 15

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PAGE 17

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Article Error You may need to use an article before this word.

PAGE 18

Missing "," Review the rules for using punctuation marks.

Prep. You may be using the wrong preposition.

P/V You have used the passive voice in this sentence. You may want to revise it using the active voice.

Missing "," Review the rules for using punctuation marks.

Missing "," Review the rules for using punctuation marks.

PAGE 19

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Missing "," Review the rules for using punctuation marks.

Article Error You may need to use an article before this word.

PAGE 20

Wrong Article You may have used the wrong article or pronoun. Proofread the sentence to make sure that the article or pronoun agrees with the word it describes.

Sp. This word is misspelled. Use a dictionary or spellchecker when you proofread your work.

Wrong Article You may have used the wrong article or pronoun. Proofread the sentence to make sure that the article or pronoun agrees with the word it describes.

PAGE 21

PAGE 22

Sentence Cap. Review the rules for capitalization.

S/V This subject and verb may not agree. Proofread the sentence to make sure the subject agrees with the verb.

Frag. This sentence may be a fragment or may have incorrect punctuation. Proofread the sentence to be sure that it has correct punctuation and that it has an independent clause with a complete subject and predicate.

Proofread This part of the sentence contains an error or misspelling that makes your meaning unclear.

PAGE 23

Missing "," Review the rules for using punctuation marks.

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